Visual System

Structure, Visual Pathways, and Reflexes

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Learning Objectives (2026)

1. Describe the structure and optical properties of the eye and relate them to image formation. (CLO 1, CLO 2)
2. Explain the organization of the retina and visual pathway from photoreceptors to the visual cortex, and correlate lesions along the pathway with characteristic visual field deficits. (CLO 1, CLO 2, CLO 3)
3. Correlate the neural pathways underlying visual reflexes with normal ocular function and common neurological disorders. (CLO 2, CLO 3)

Pre-Session Assignment

- Preview the PowerPoint slides. You are not expected to memorize details before class.
- Boards & Beyond B&B: Eye Anatomy (26 min)
- B&B: Retina (14 min)
- B&B: Visual Field Defects (13 min)

Eye Structure & Image Formation

Layers of the Eyeball

Fibrous (Outer) Layer

- Sclera: protection and extraocular muscle attachment
- Cornea: transparent, major refractive surface

Vascular (Middle) Layer — Uvea

- Choroid: vascular supply to outer retina
- Ciliary body: aqueous humor production and accommodation
- Iris: regulates pupil size

Neural (Inner) Layer

- Retina: detects light and initiates visual signaling

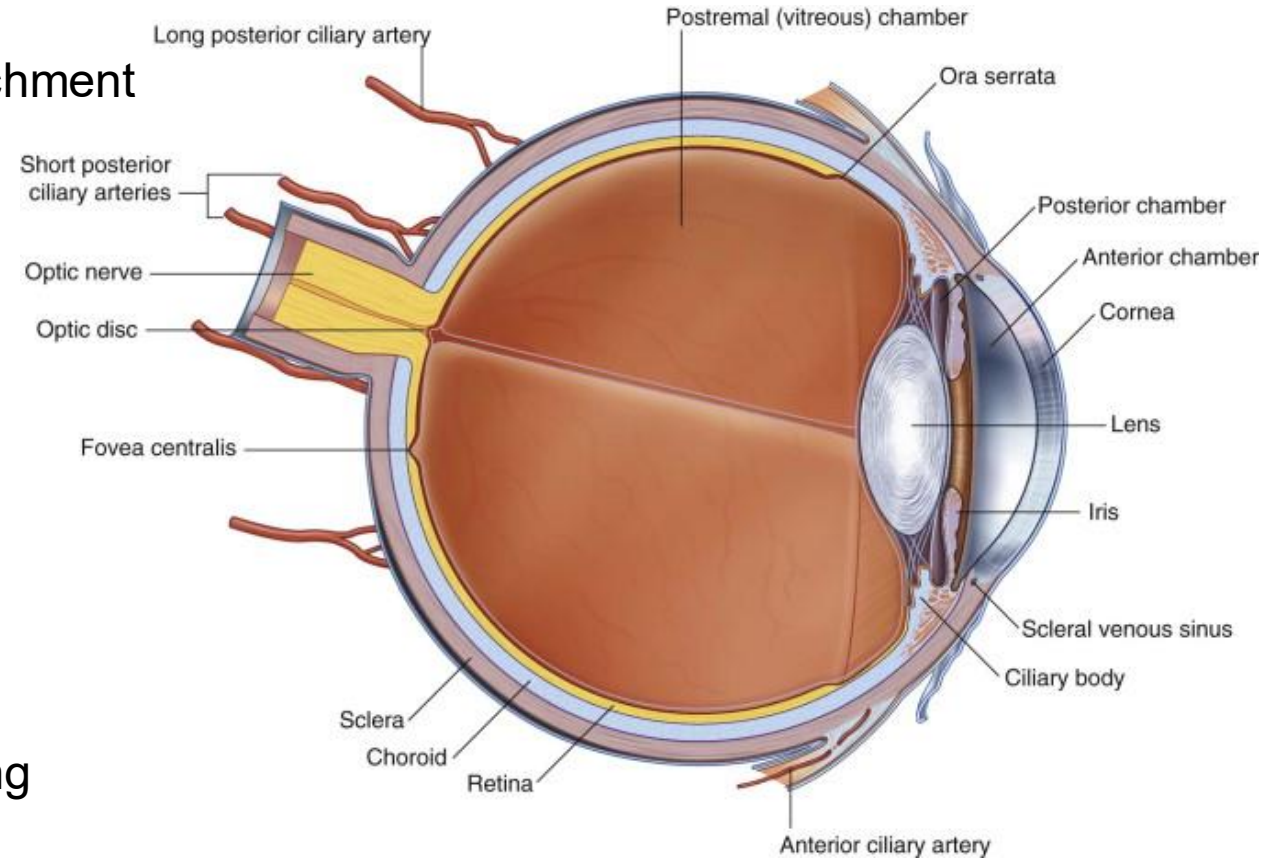
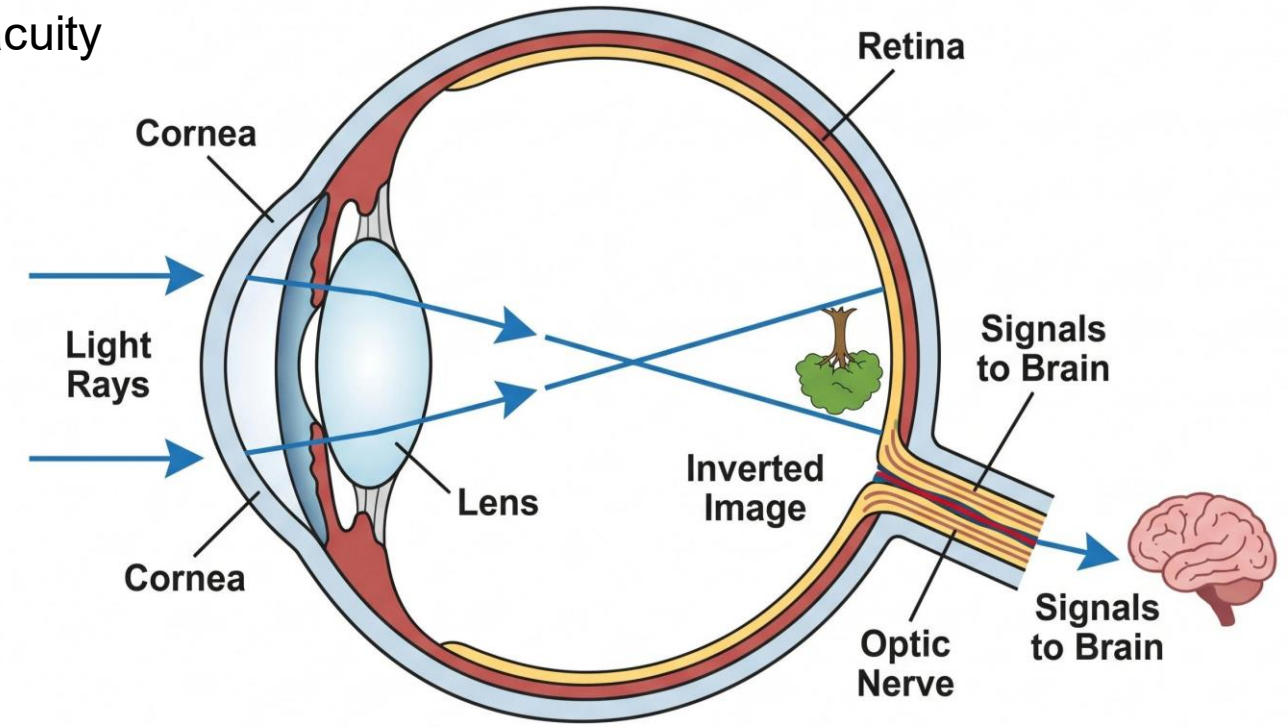


Fig. 8.105 Gray's Anatomy for Students, 3e.

Fibrous = protection/refraction | Vascular = nutrition/control | Neural = vision

Cornea, Lens, Image Formation

- **Cornea:** provides approximately two-thirds of the eye's refractive power
- **Lens:** adjusts refractive power for near and distant vision
- Light rays converge on the retina, forming an inverted and reversed image
- **Fovea centralis:** region of highest visual acuity



Cornea = major refraction | Lens = adjustable focus | Retina = image detection

Ciliary Body & Accommodation

Distant Vision

- Ciliary muscle relaxes → zonular fibers tighten
- Lens flattens → lower refractive power

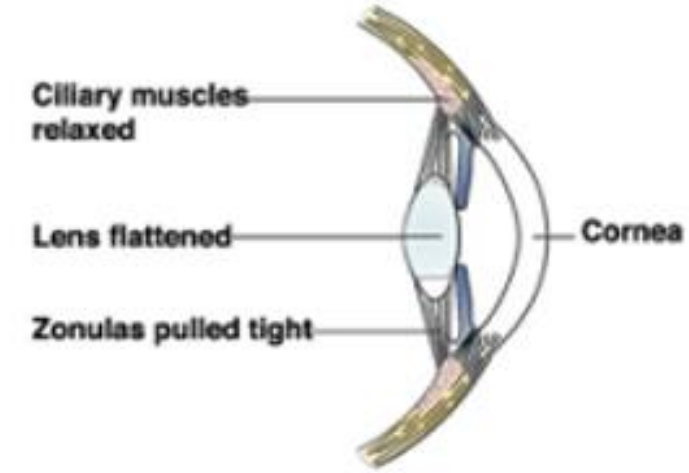
Near Vision

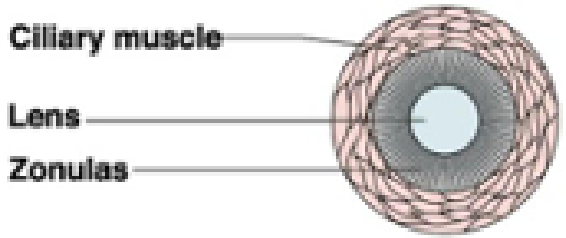
- Ciliary muscle contracts → zonular fibers loosen
- Lens becomes rounder → higher refractive power

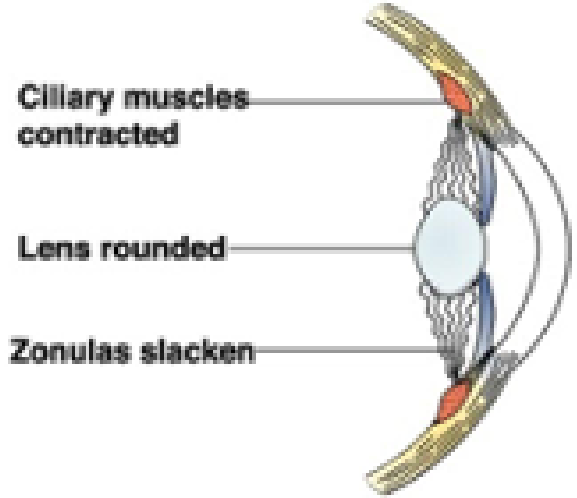
Innervation

- Parasympathetic: CN III → ciliary ganglion → short ciliary nerves

Near vision: ciliary muscle contracts → zonules relax → lens rounds







Iris & Pupillary Control

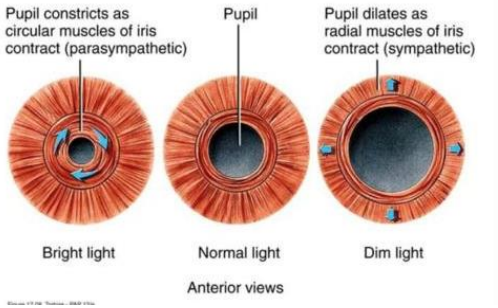
Iris: pigmented diaphragm controlling pupil size

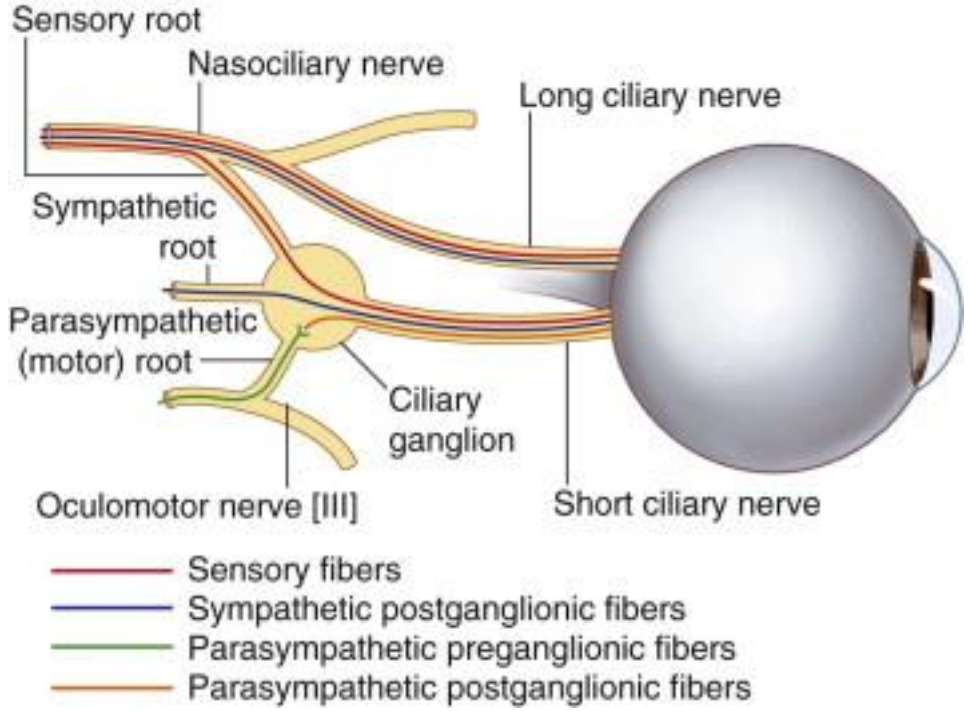
Sphincter Pupillae

- Parasympathetic: CN III → ciliary ganglion → short ciliary nerves
- Contraction → miosis (pupil constriction)

Dilator Pupillae

- Sympathetic: superior cervical ganglion → long ciliary nerves (principal route)
- Contraction → mydriasis (pupil dilation)





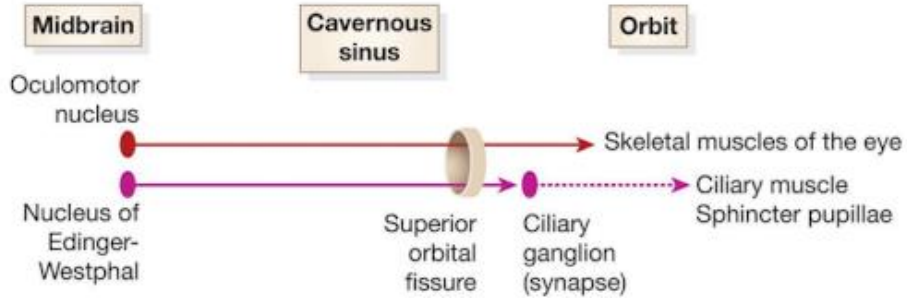


Fig. 8.103 Gray's Anatomy for Students, 3e. . Champney. ECNA, 2016.

Retina, Macula, Fovea & Optic Disc

Retina

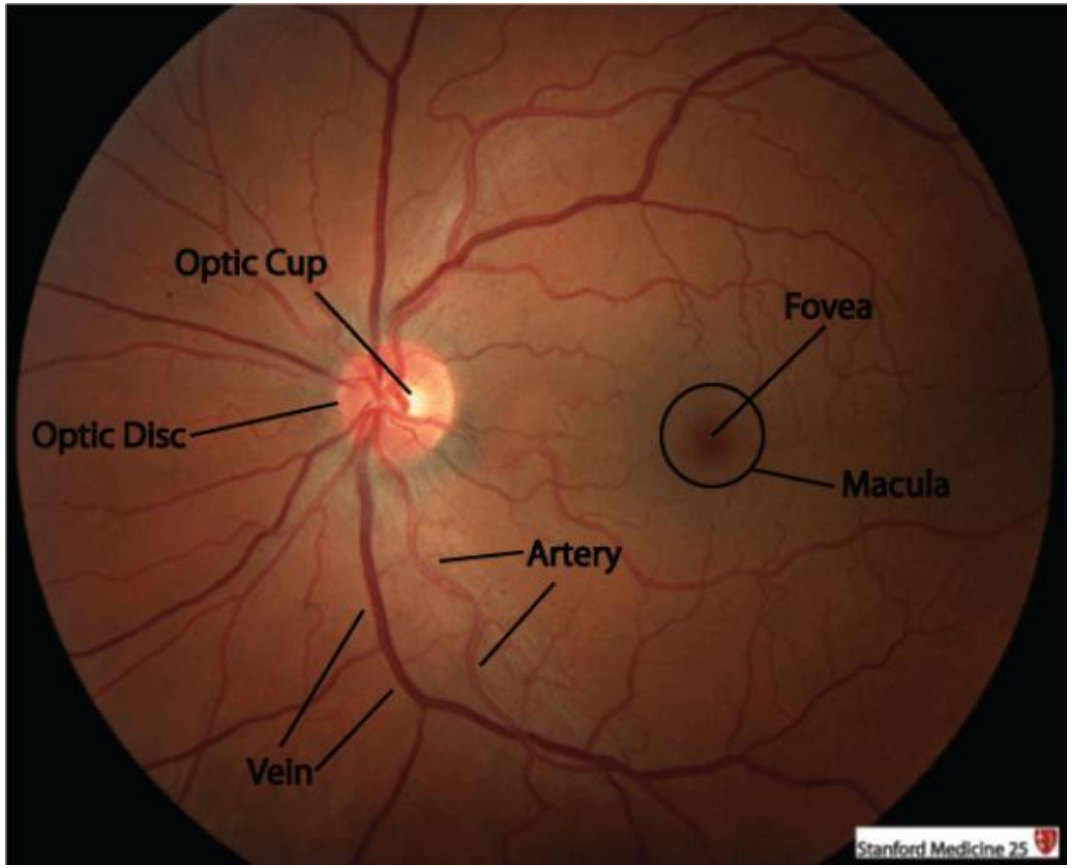
- Neural layer responsible for light detection
- Ganglion-cell axons form the optic nerve (CN II)

Macula Lutea

- Central retinal region responsible for detailed vision
- Fovea centralis: highest cone density and visual acuity

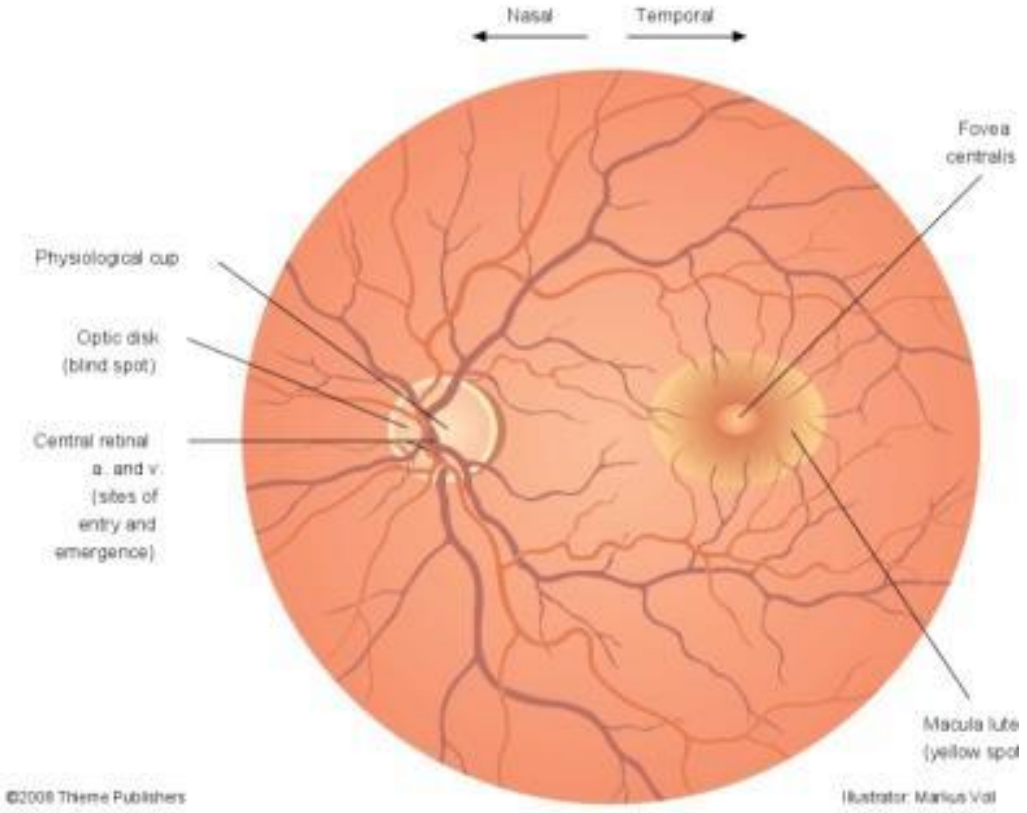
Optic Disc

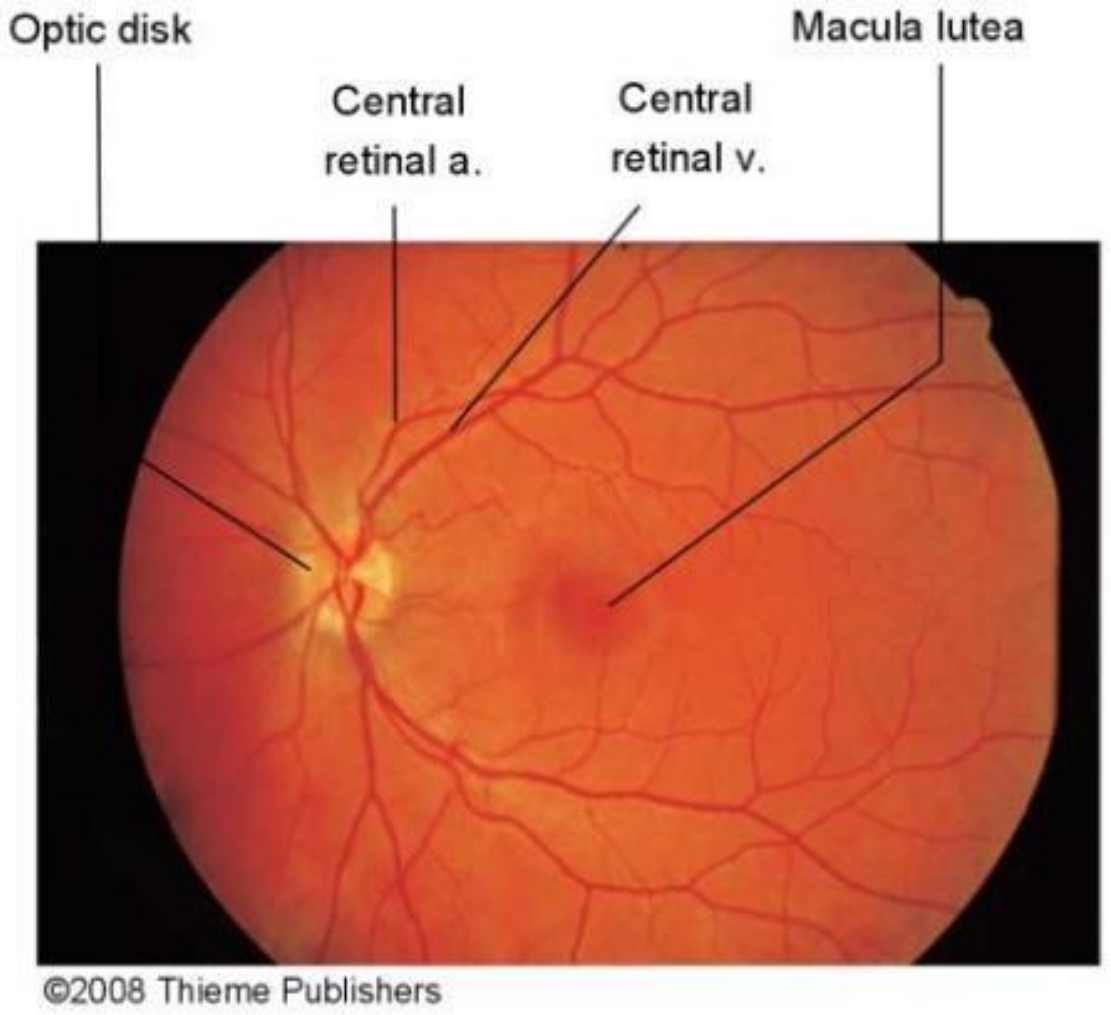
- Exit of optic nerve and central retinal vessels
- No photoreceptors → physiological blind spot



Fovea = highest visual acuity | Optic disc = blind spot

Fundus


Nasal
Temporal
Fovea centralis
Physiological cup
Optic disk (blind spot)
Central retinal a. and v. (sites of entry and emergence)
Macula lutea (yellow spot
©2008 Thieme Publishers
Illustrator: Markus Voll


Optic disk
Central retinal a.
Central retinal v.
Macula lutea
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Papilledema



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Normal



Papilledema

Visual Fields & Visual Pathways

Each cerebral hemisphere processes information from the opposite visual hemifield.

Visual Fields vs Retinal Fields

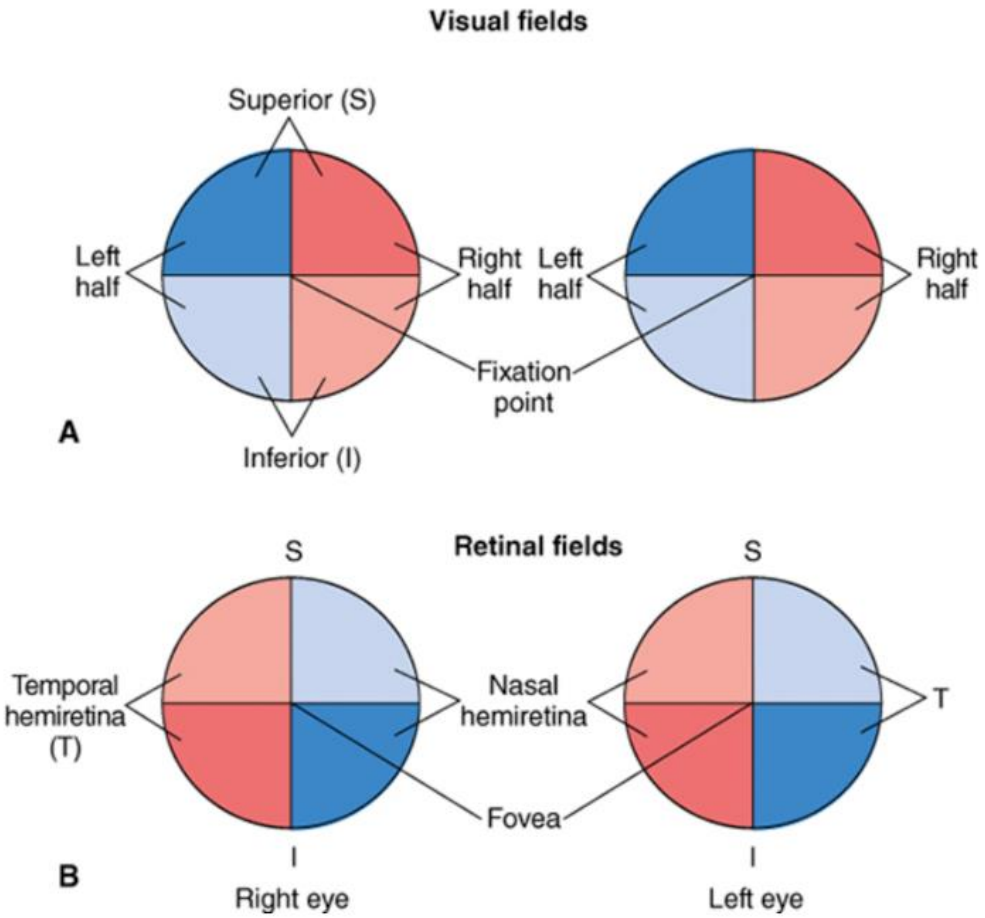
Visual Field

- Area of space visible while looking straight ahead
- Divided into right/left and superior/inferior visual fields

Retinal Field

- Image projected onto the retina is inverted and reversed
- Left visual field → right half of each retina
- Right visual field → left half of each retina
- Superior visual field → inferior retina

Visual field and retinal location are opposite

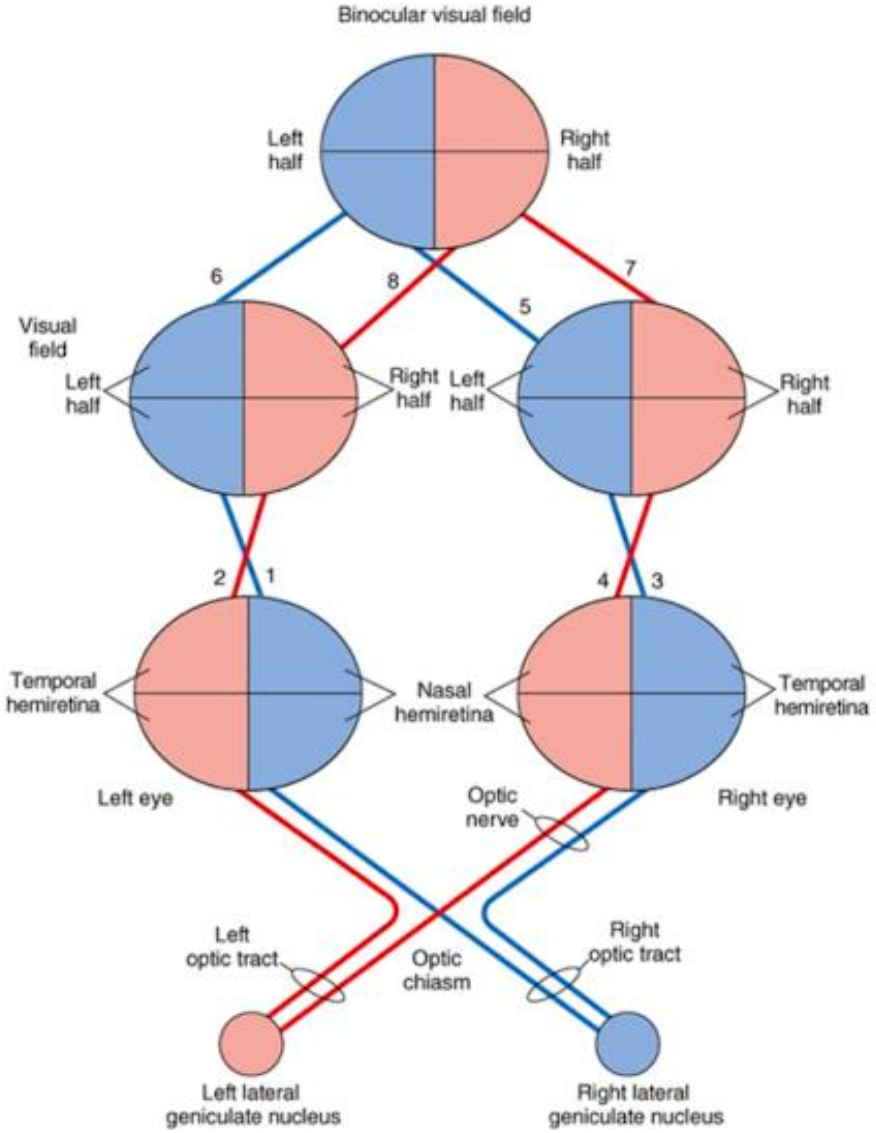


Siegel, *Essential Neuroscience*

Nasal vs Temporal Hemiretina

- Each retina is divided into nasal and temporal hemiretinae
- Temporal visual field → nasal retina
- Nasal visual field → temporal retina
- Each eye receives information from both visual hemifields
- Binocular visual fields overlap centrally

Temporal visual field → nasal retina
Nasal visual field → temporal retina

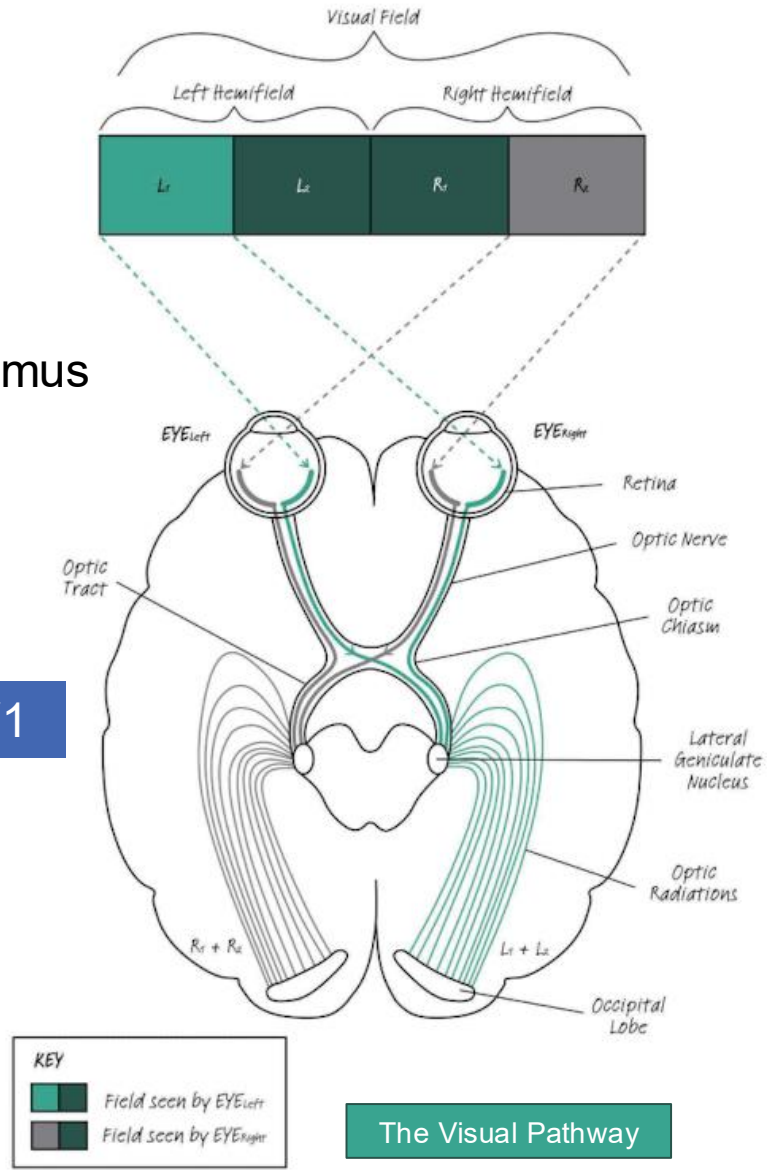


Siegel, *Essential Neuroscience*

Visual Pathway: Retina to Cortex

- Retinal ganglion-cell axons form the optic nerve (CN II)
- Optic nerves converge at the optic chiasm
- Fibers continue as optic tracts
- Most fibers synapse in the lateral geniculate nucleus (LGN) of the thalamus
- Optic radiations project to the primary visual cortex (V1)

Retina → CN II → Chiasm → Optic tract → LGN → Optic radiations → V1



The Visual Pathway

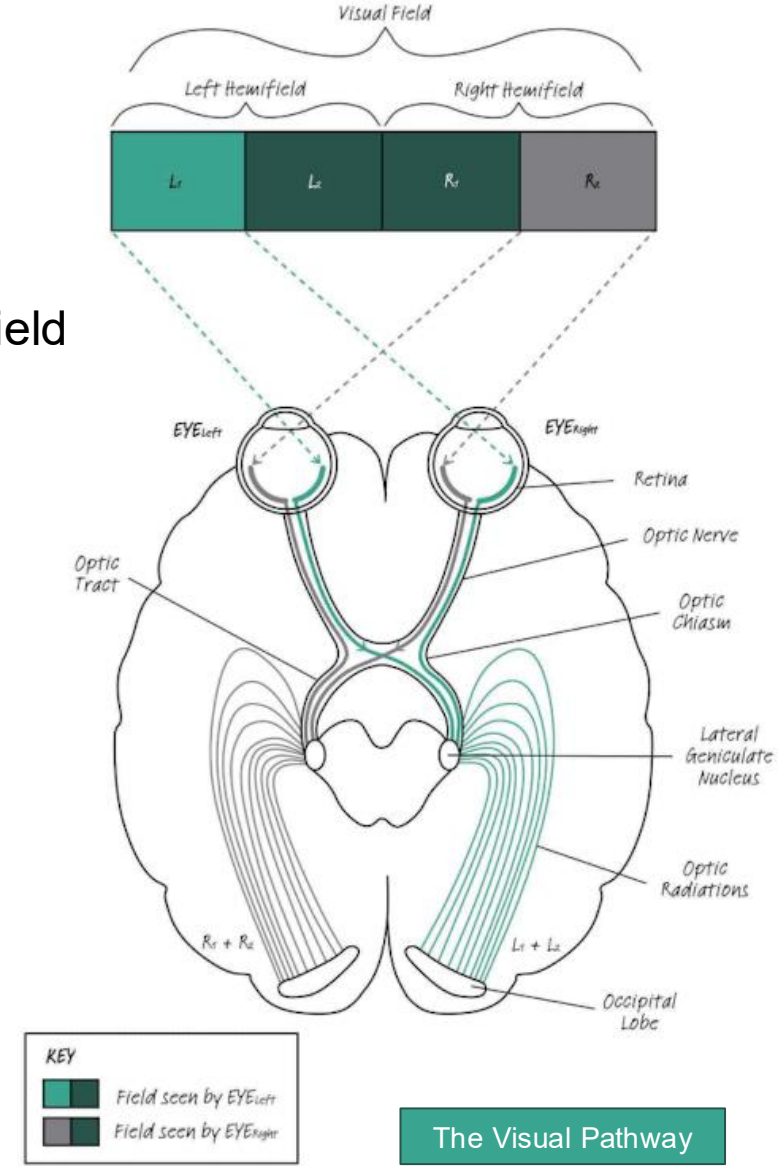
Berger. CCNA

Optic Chiasm: Which Fibers Cross?

- Nasal retinal fibers cross at the optic chiasm
- Temporal retinal fibers remain ipsilateral
- Each optic tract carries information from the contralateral visual hemifield
- Right optic tract → left visual field
- Left optic tract → right visual field

Nasal fibers CROSS | Temporal fibers STAY

Which visual hemifield is carried by the right optic tract?



The Visual Pathway

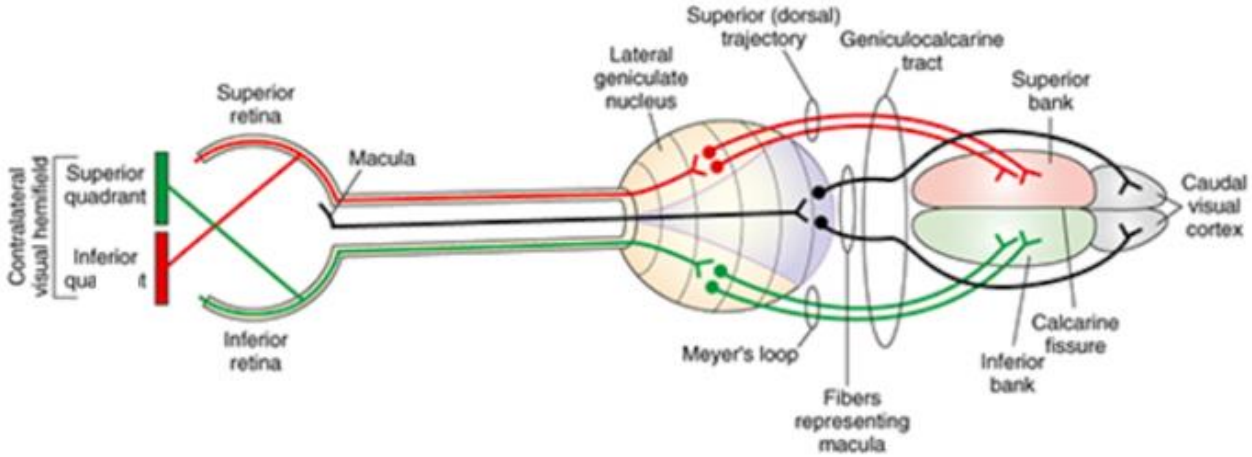
Lateral Geniculate Nucleus & Optic Radiations

Lateral Geniculate Nucleus (LGN)

- Visual relay nucleus of the thalamus
- Receives input from the contralateral visual hemifield
- Projects to the primary visual cortex via optic radiations

Optic Radiations

- Temporal lobe (Meyer's loop): inferior retinal fibers → superior visual field
- Parietal lobe: superior retinal fibers → inferior visual field



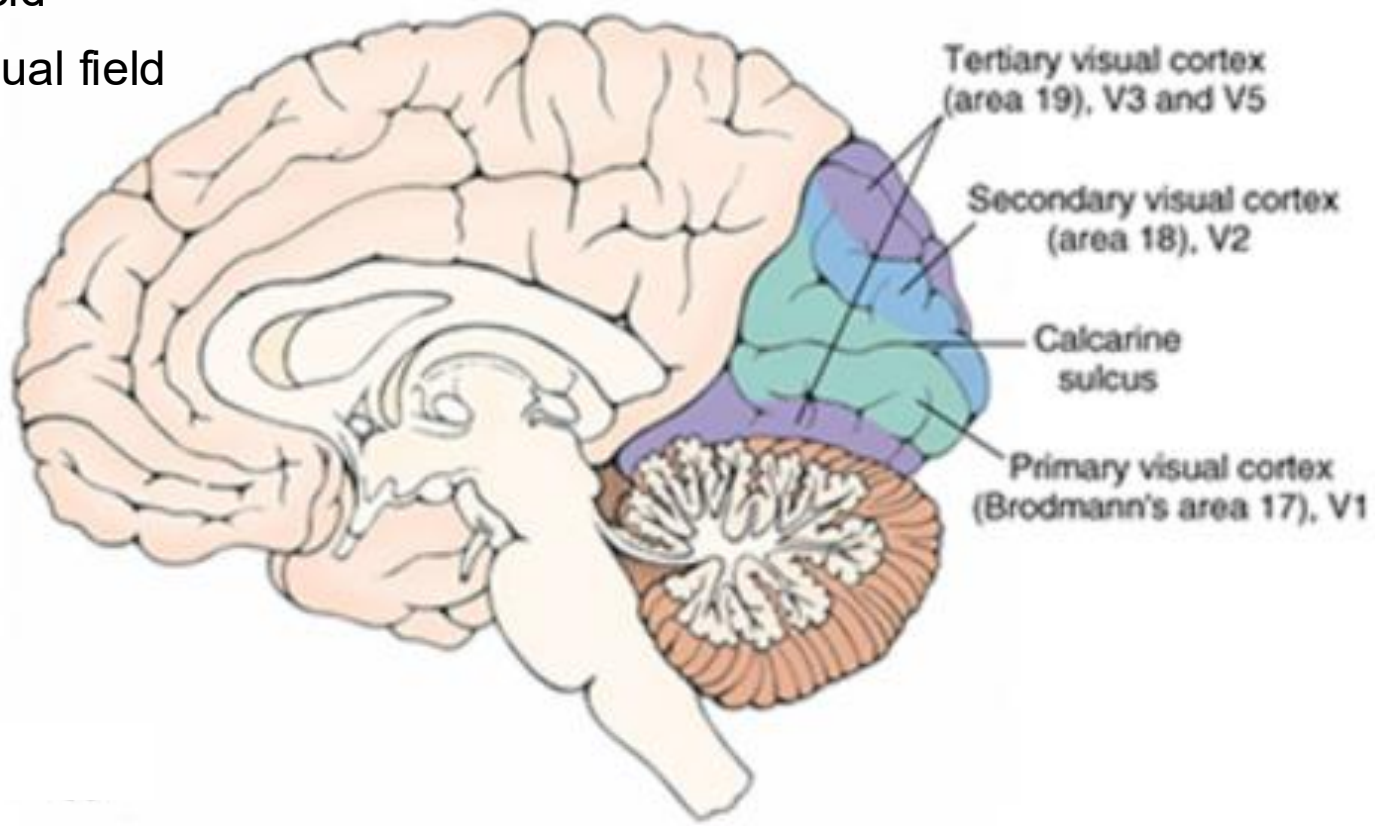
Meyer's loop → superior visual field

Parietal radiations → inferior visual field

Siegel, *Essential Neuroscience*

Primary Visual Cortex (V1)

- Located along the calcarine sulcus of the occipital lobe
- Cuneus (superior bank): inferior visual field
- Lingual gyrus (inferior bank): superior visual field
- Occipital pole: central (macular) vision
- More anterior visual cortex represents peripheral vision



Cuneus = inferior visual field
Lingual gyrus = superior visual field

Siegel, *Essential Neuroscience*

Visual Pathway: Three Rules for Localization

Rule 1 — Retinal Projection

- Visual fields project to the opposite retinal halves

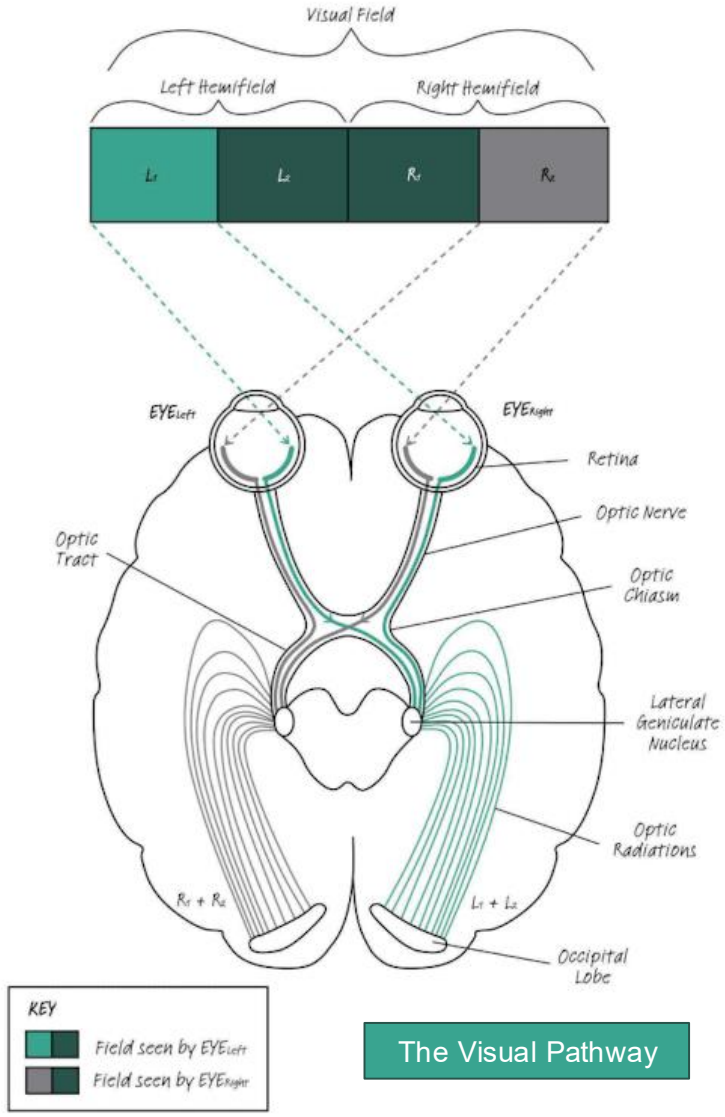
Rule 2 — Optic Chiasm

- Nasal retinal fibers cross; temporal retinal fibers remain ipsilateral

Rule 3 — Cerebral Representation

- Each hemisphere processes the contralateral visual hemifield
- Temporal radiations carry superior-field information
- Parietal radiations carry inferior-field information

Visual field → Retina → CN II → Chiasm →
Optic tract → LGN → Optic radiations → Cortex



The Visual Pathway

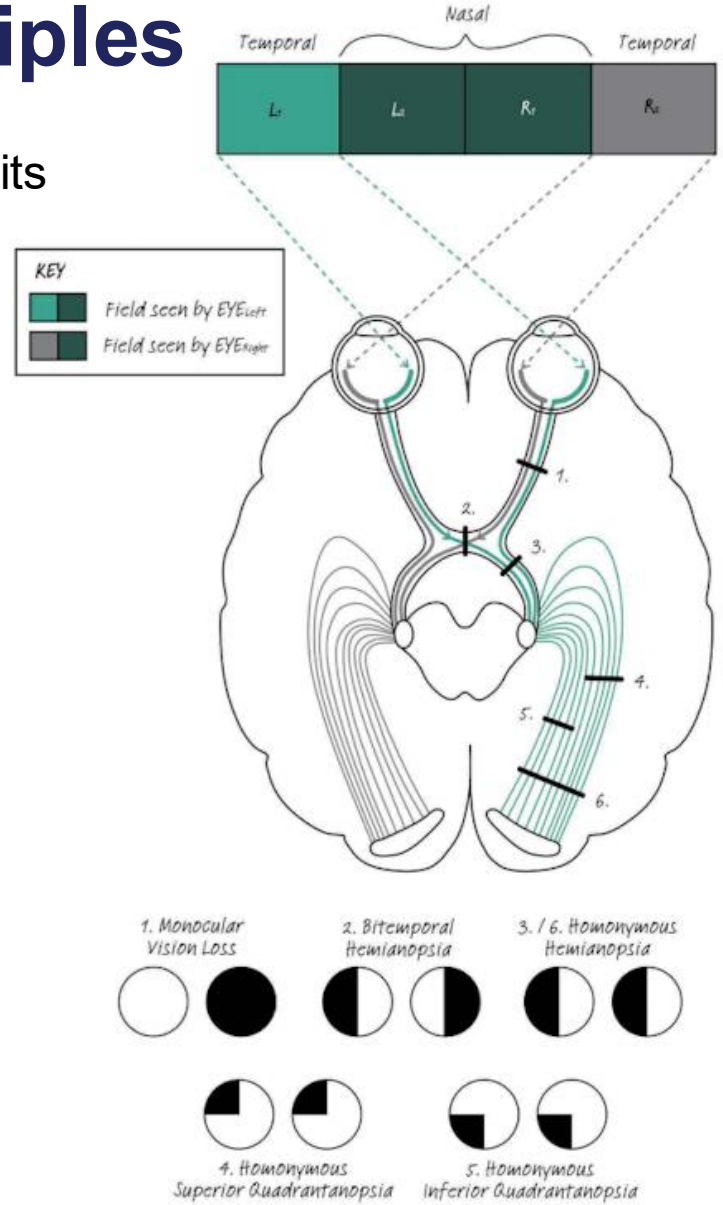
Berger. CCNA

Visual Field Defects & Lesion Localization

Visual Field Defects: Localization Principles

- Lesions along the visual pathway produce characteristic visual-field deficits
- **Prechiasmal lesions:** monocular visual loss
- **Chiasmal lesions:** heteronymous visual-field defects
- **Postchiasmal lesions:** contralateral homonymous visual-field defects
- Lesion location determines the pattern of visual loss

Before chiasm → monocular
Chiasm → heteronymous
After chiasm → homonymous



Visual Field Defects

Berger. CCNA

Optic Nerve & Optic Chiasm Lesions

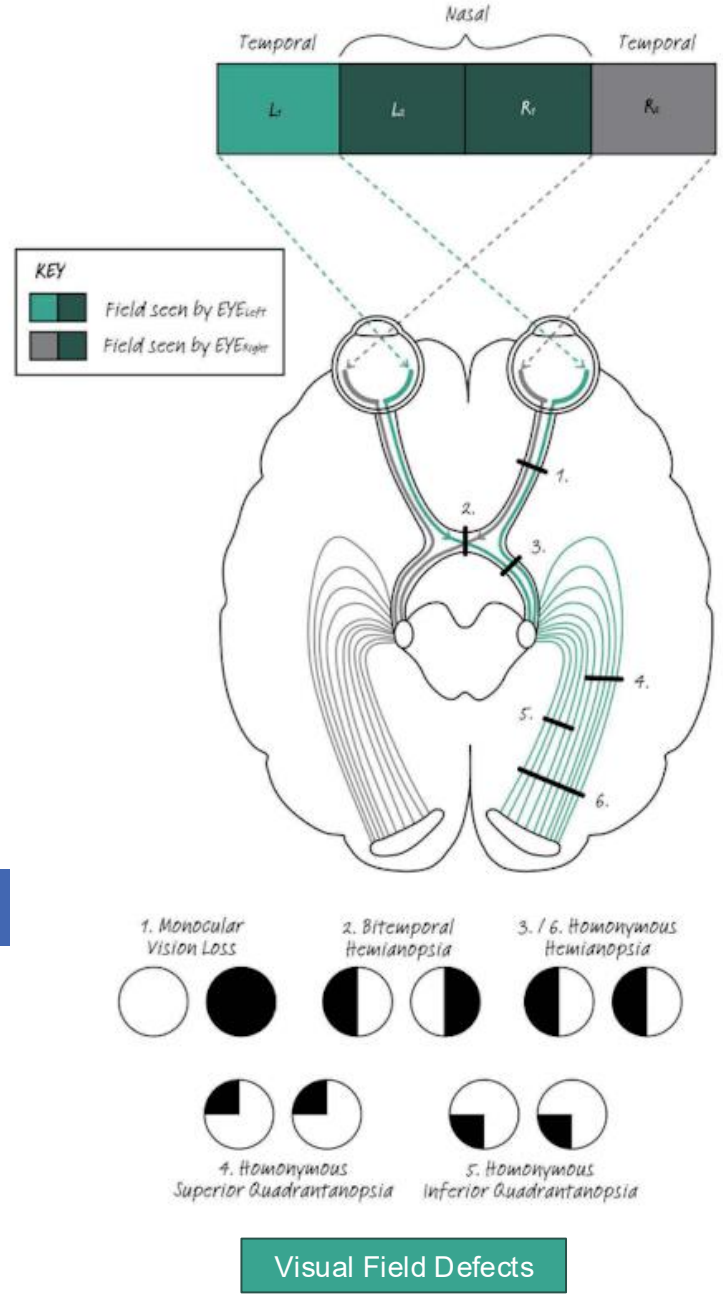
Optic Nerve Lesion

- Ipsilateral monocular visual loss
- Loss of visual input from the affected eye

Optic Chiasm Lesion

- Damage to crossing nasal retinal fibers
- Bitemporal hemianopia
- Classically associated with pituitary mass compression

Optic nerve → monocular loss | Optic chiasm → bitemporal hemianopia



Visual Field Defects

Berger. CCNA

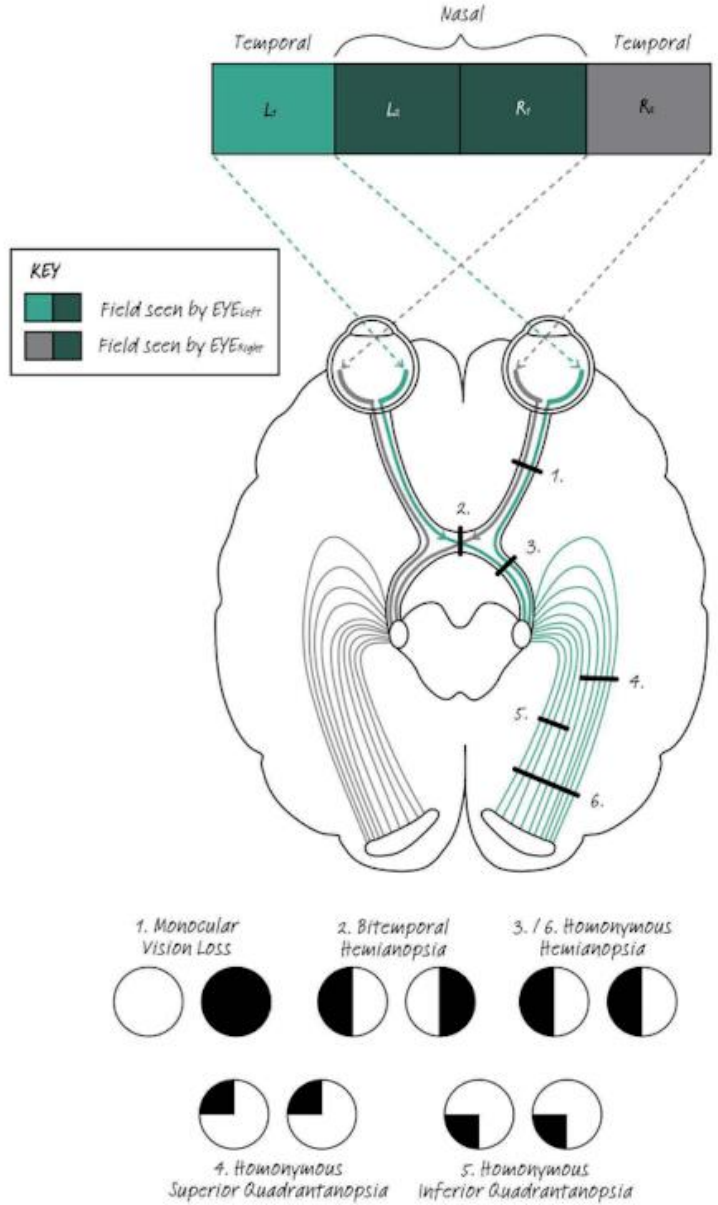
Optic Tract Lesion

- Each optic tract carries information from the contralateral visual hemifield
- Lesion produces contralateral homonymous hemianopia
- Both eyes lose the same half of their visual fields

Example

- Right optic tract lesion → left homonymous hemianopia

Right optic tract lesion → left homonymous hemianopia



Visual Field Defects

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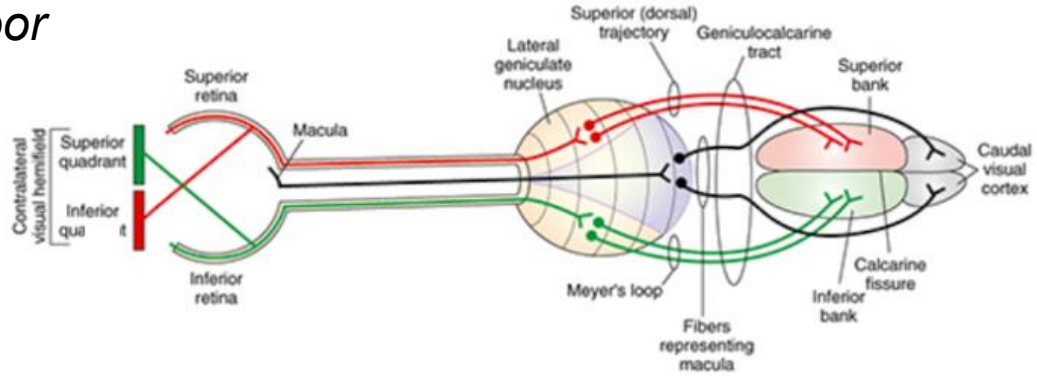
Optic Radiations: Temporal vs Parietal Lesions

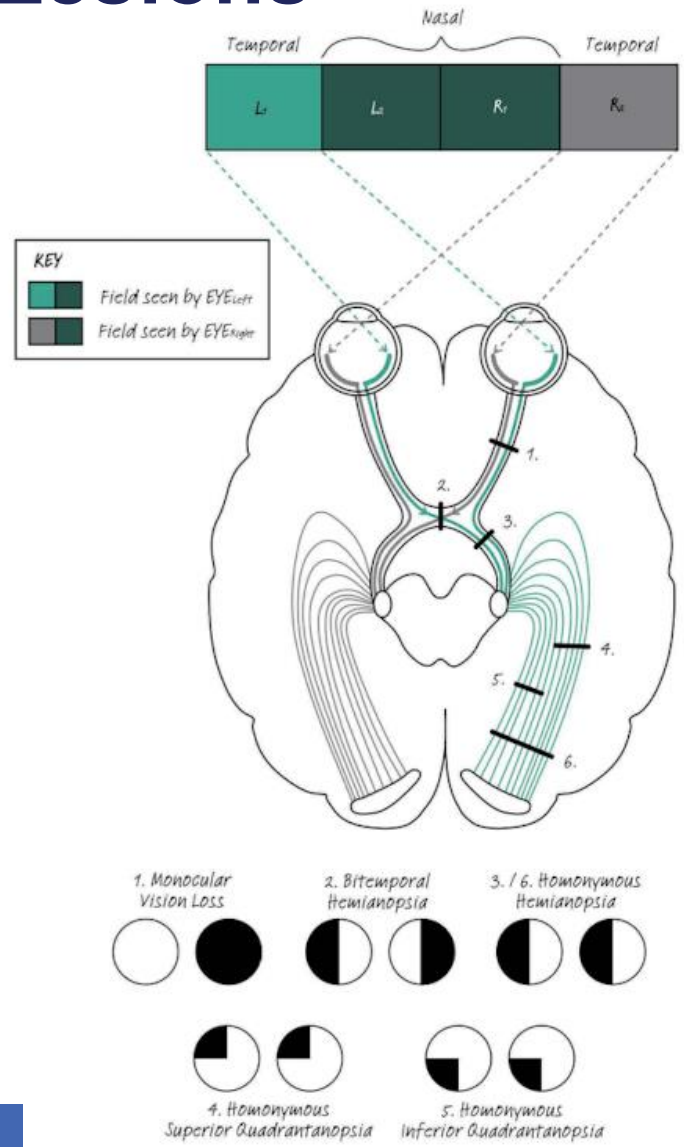
Temporal Lobe — Meyer's Loop

- Carries information from the contralateral superior visual field
- Lesion → contralateral superior quadrantanopia
- *Pie in the sky*

Parietal Lobe — Superior Optic Radiations

- Carries information from the contralateral inferior visual field
- Lesion → contralateral inferior quadrantanopia
- *Pie on the floor*





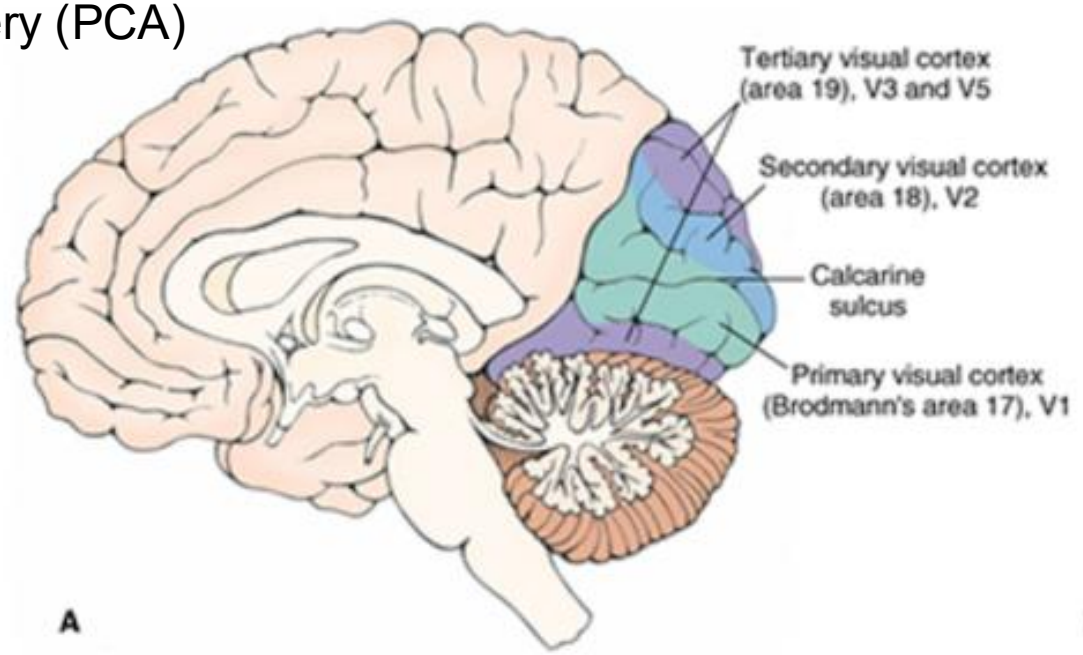
Temporal → superior field loss | Parietal → inferior field loss

Visual Field Defects

Siegel, Berger

Primary Visual Cortex Lesions

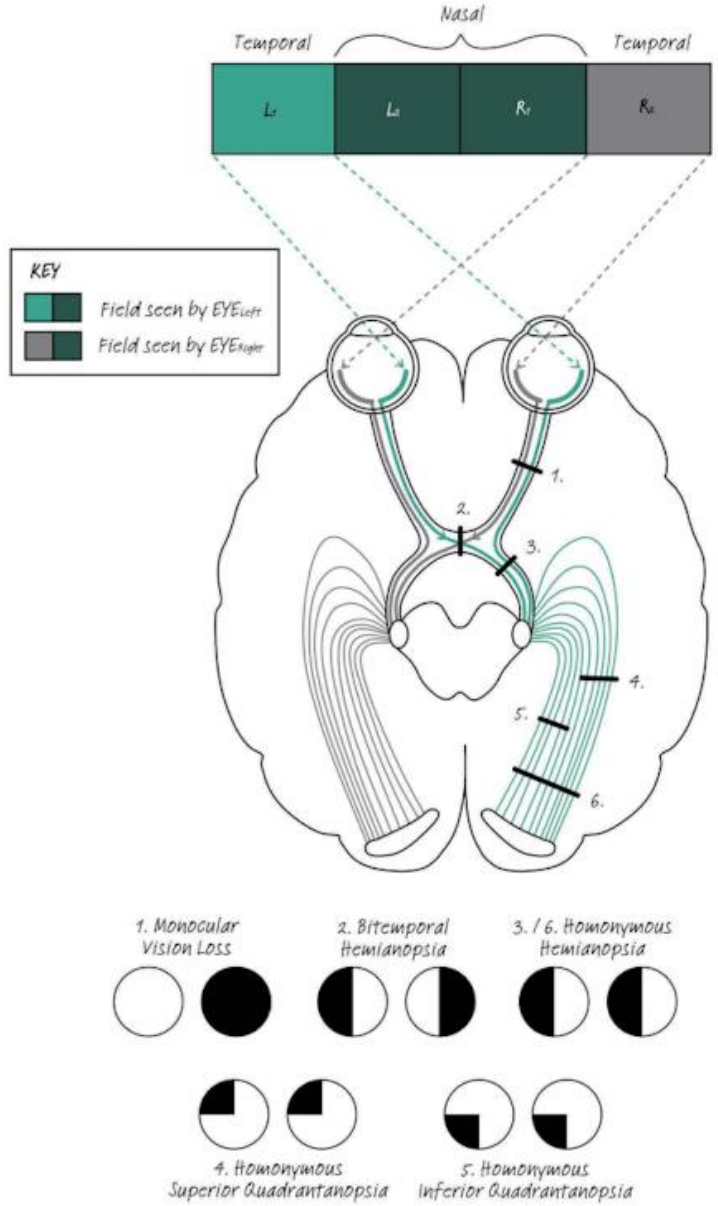
- Primary visual cortex (V1) lies along the calcarine sulcus
- Unilateral lesion → contralateral homonymous hemianopia
- Occipital pole: represents central (macular) vision
- Some occipital lesions produce macular sparing
- Common vascular association: posterior cerebral artery (PCA)



Siegel, *Essential Neuroscience*

Visual Field Defects: Summary

Lesion Site	Visual Field Deficit
Optic nerve	Ipsilateral monocular visual loss
Optic chiasm	Bitemporal hemianopia
Optic tract	Contralateral homonymous hemianopia
Temporal optic radiations	Contralateral superior quadrantanopia
Parietal optic radiations	Contralateral inferior quadrantanopia
Primary visual cortex	Contralateral homonymous hemianopia ± macular sparing



Visual Field Defects

Berger. CCNA

Visual System:

Structure, Visual Pathways, and Reflexes

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